

Helix OTA “Issues (open workable items)

Revision: 2 Last modified: 2026-06-10T18:30:00Z

This is the canonical open-work tracker (Â§11.4.15 Status, Â§11.4.16 Type, Â§11.4.54 ATM-NNN). Closed items migrate to [Fixed.md](#). The short-form companion is [Issues_Summary.md](#). Items surfaced during the 2026-06-10 emulator-driven device-testing session; HEAD at creation a839220 + the docs commit on top.

Â§3. [ATM-003] Emulator Tier-2 “real Android A/B (update_engine/AVB/dm-verity auto-rollback) is host-gated

Status: Operator-blocked Type: Task

Operator-Block-Details: - **WHAT:** Stand up the Tier-2 emulator “a real Android A/B update_engine payload-apply with AVB/dm-verity verification and auto-rollback, driven end-to-end against the control plane (per docs/design/EMULATED_DEVICE_TESTING.md). - **WHY:** Tier-2 requires Cuttlefish (cvd) on a Linux host with nested KVM. Self-resolution exhausted (Â§11.4.21): (a) no host CLI/virtualisation path “the current host is Apple-Silicon with applehv, which cannot run Cuttlefish or nested KVM; (b) subagent delegation cannot conjure the missing kernel virtualisation; (c) the containers submodule (podman/applehv) cannot host a KVM-backed Cuttlefish guest on this hardware; (d) no captured fallback substitutes for real update_engine partition behaviour; (e) external research confirms Cuttlefish“s Linux+KVM requirement “not a tooling gap. NOT structurally impossible (Â§11.4.112) “host/hardware-gated only. - **UNBLOCK CONDITION:** Access to a Linux host (or CI runner) with nested-KVM enabled where Cuttlefish (cvd) boots. - **WHO:** Operator “provision the Linux+KVM host / CI runner; see docs/design/EMULATED_DEVICE_TESTING.md Tier-2.

Â§4. [ATM-004] Emulator Tier-3 “real RK3588 / Orange Pi 5 Max vendor HAL, U-Boot slot-switch, dm-verity on real partitions

Status: Operator-blocked Type: Task

Operator-Block-Details: - **WHAT:** Tier-3 validation on the physical board “real vendor HAL, U-Boot A/B slot-switch, and dm-verity over real partitions, exactly as a fielded device runs (per docs/design/EMULATED_DEVICE_TESTING.md). - **WHY:** Requires the physical RK3588 / Orange Pi 5 Max hardware, which is not attached to this host. Self-resolution exhausted (Â§11.4.21): (a) no remote board over the available CLIs/ADB/SSH; (b) subagent delegation cannot substitute for absent silicon; (c) no repo tooling emulates real U-Boot slot-switch + on-silicon dm-verity; (d) no captured fallback reproduces real-partition AVB rollback; (e) research confirms the vendor HAL + bootloader path is board-specific. NOT structurally impossible (Â§11.4.112) “hardware-gated only. - **UNBLOCK CONDITION:** A physical RK3588 / Orange Pi 5 Max board reachable over ADB/SSH (or physically attached) for flashing. - **WHO:** Operator “attach / provide remote access to the board.